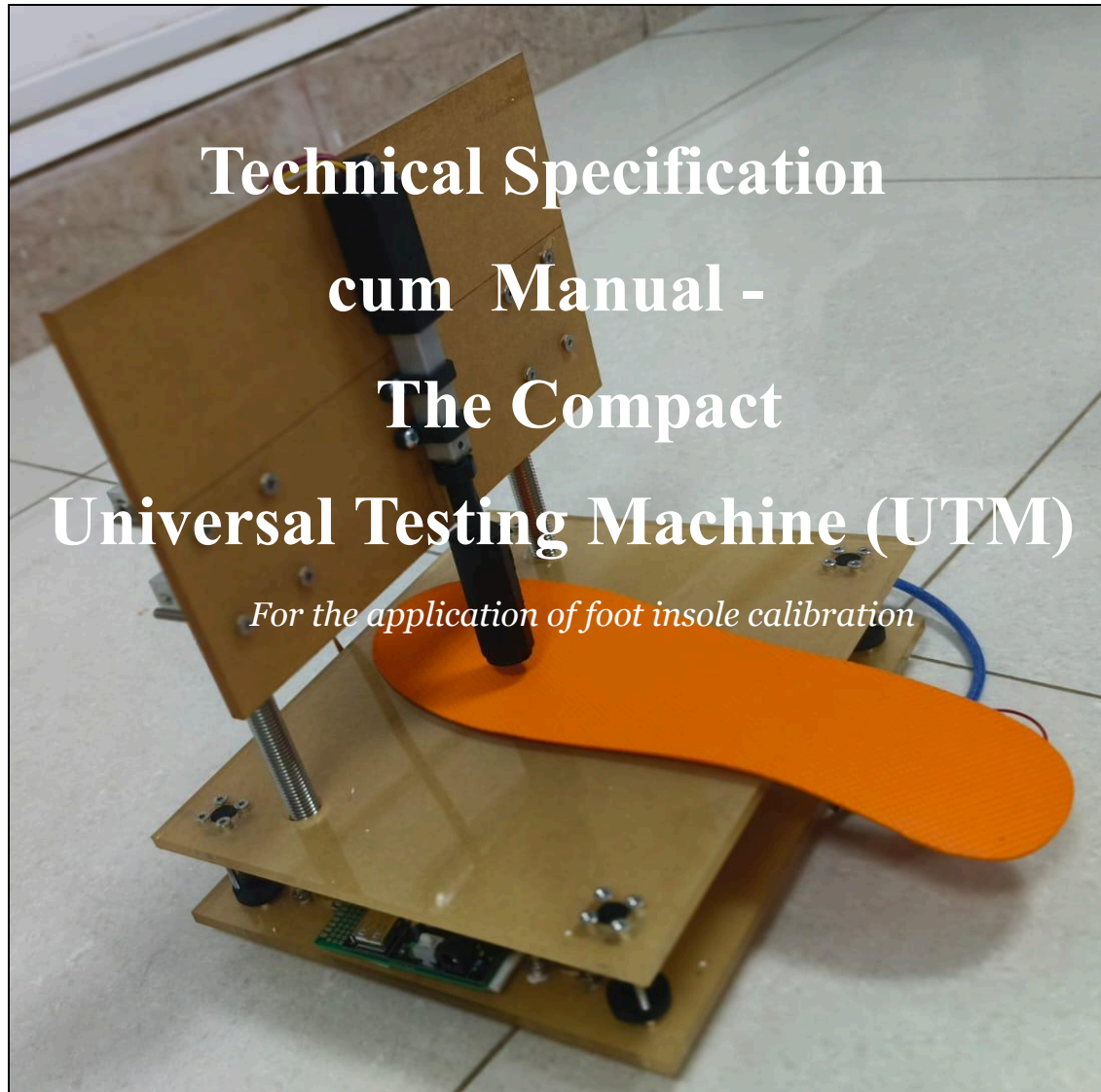




CoE-AH

CENTER OF EXCELLENCE IN
AFFORDABLE HEALTHCARE



Technical Specification cum Manual - The Compact Universal Testing Machine (UTM)

For the application of foot insole calibration

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About Us-

The Center of Excellence in Affordable Healthcare (CoE-AH) at IIT Kharagpur is devoted to revolutionizing healthcare accessibility, availability, and efficiency. By bridging the intersections of advanced technology, medical research, and interdisciplinary collaborations, CoE-AH aspires to deliver innovative solutions that ensure quality healthcare is within everyone's reach. Emphasizing sustainable and cost-effective practices, our mission is to lay the foundation for a healthier future, reinforcing our commitment to the global imperative of affordable and inclusive medical care.

Our Mission-

- Conducting multidisciplinary applied research focused on developing affordable and inclusive healthcare technologies with global standards
- Developing patient-centric healthcare technologies that are effective, affordable, sustainable, and accessible to improve overall well-being and quality of life.
- Bridging the gap between advanced technology, medical research, and interdisciplinary collaborations

Our Vision-

- Lead transformative healthcare by combining advanced technology and innovative research for accessibility.
- Redefine global healthcare through sustainable, affordable, and inclusive practices.
- Set the benchmark in affordable healthcare innovations, promoting health quality and access.
- Drive innovation and collaboration to ensure advanced medical solutions are university accessible

Introduction-

A Universal Testing Machine (UTM) is a versatile instrument widely used to evaluate the mechanical properties of materials, components, and structures. Traditional UTMs are typically designed to apply large forces and determine the maximum load a material can withstand through various tests, including tensile (pulling), compression (pushing), bending, shear, hardness, and torsion.

However, standard UTMs often lack the precision required to apply lower forces in the range of 5N to 120N. This limitation presents a significant challenge for applications that demand accurate low-force measurements.

The primary objective behind developing our compact UTM is to enable precise calibration of foot insoles used in GAIT analysis—an essential process for measuring the force distribution exerted by the foot at various points. The sensors embedded within these insoles must be calibrated by applying varying forces within the 5N to 120N range to ensure accurate readings.

To address this need, we have designed a compact UTM system that offers precise force control specifically tailored for sensor calibration. This innovation ensures reliable and repeatable calibration of insole sensors, supporting accurate GAIT analysis and enhancing the overall performance and reliability of foot pressure measurement systems.



**Insole calibration
using the Compact UTM**

Overview of The Mini UTM-

Our system is specifically designed to calibrate force sensors used in foot insoles for GAIT analysis within the biomechanics domain. Unlike conventional Universal Testing Machines (UTMs), which primarily focus on determining the maximum force a material can withstand, our mini UTM is engineered for a different application.

The primary objective of our system is to apply precise and small forces, ensuring accurate calibration of force-detecting sensors. This focus on low-force application makes our system uniquely suited for the sensitive requirements of GAIT analysis, where precision and reliability in sensor calibration are essential.

Design-

Mechanical Structure-

The mechanical structure is engineered so that it can handle a force of 120N without any compliance in the structure; All the forces are precisely counteracted by opposing forces within the design. This force cancellation minimizes structural compliance, ensuring that internal stresses are evenly distributed and deformation is kept to a minimum. As a result, the structure maintains high stiffness and stability under load.

The UTM is divided into 3 parts-

1. Base
2. Body
3. Head

Base:

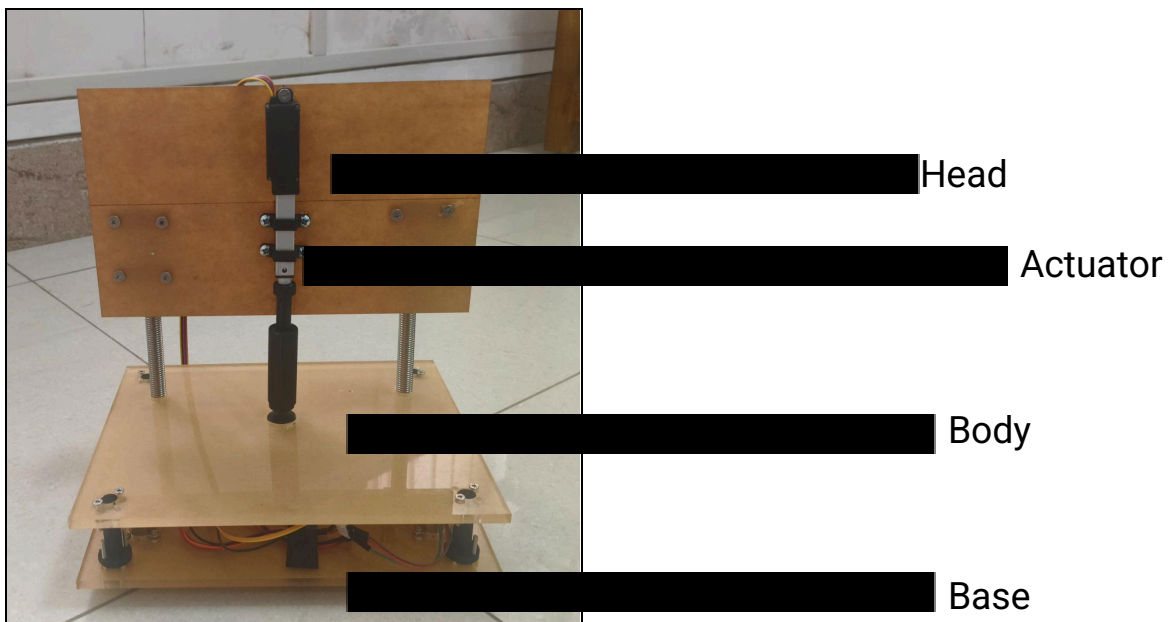
It is the foundation of the UTM, made of 6mm acrylic, gives room for all the electronic components, and stabilizes the entire system from jerking when the linear actuator is activated

Body:

It is made of 6mm acrylic, facilitating the area for sliding the Insole so making it easy to calibrate all the sensors easily.

Head:

This carries the Linear Actuator, which is used to apply force on the insole for calibration. The Head is made of 6mm acrylic and is designed in such a way that the torsional torques generated by the linear actuator counteract each other across the pillars.



Front view

Fragments of the Mini UTM



Front view of the shaft



Top view of the shaft



Bottom of the shaft



Detachable Magnetic base



Assembly of shaft and magnetic base



Cross sectional view of shaft





Base to place Load cell



Complete setup



Spacer

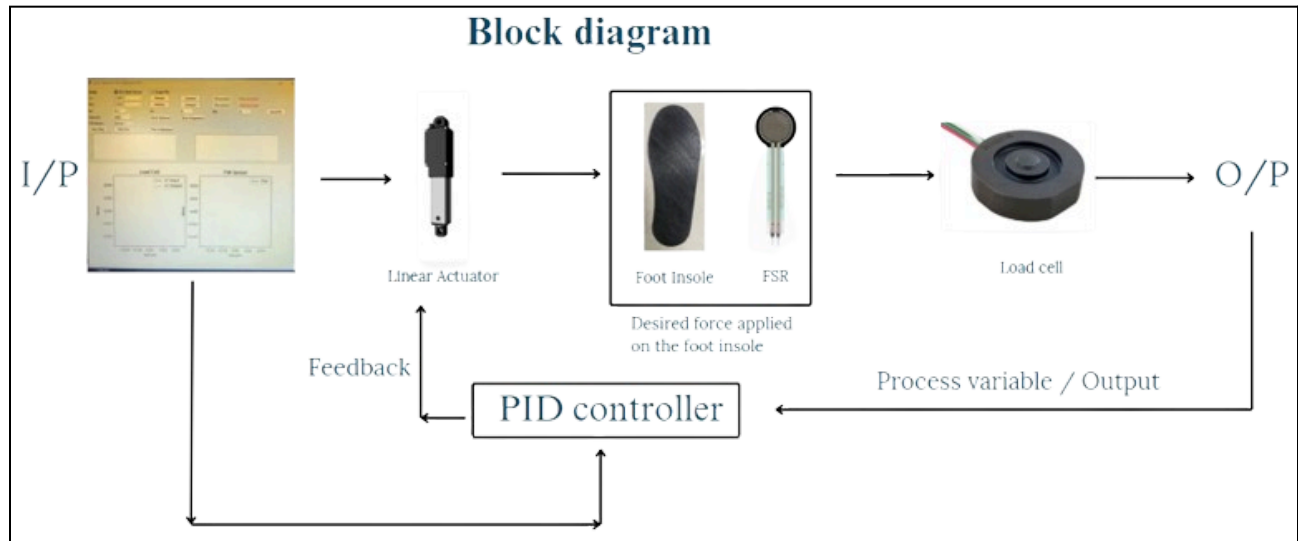


Mini UTM Assembly

Force Control Scheme-

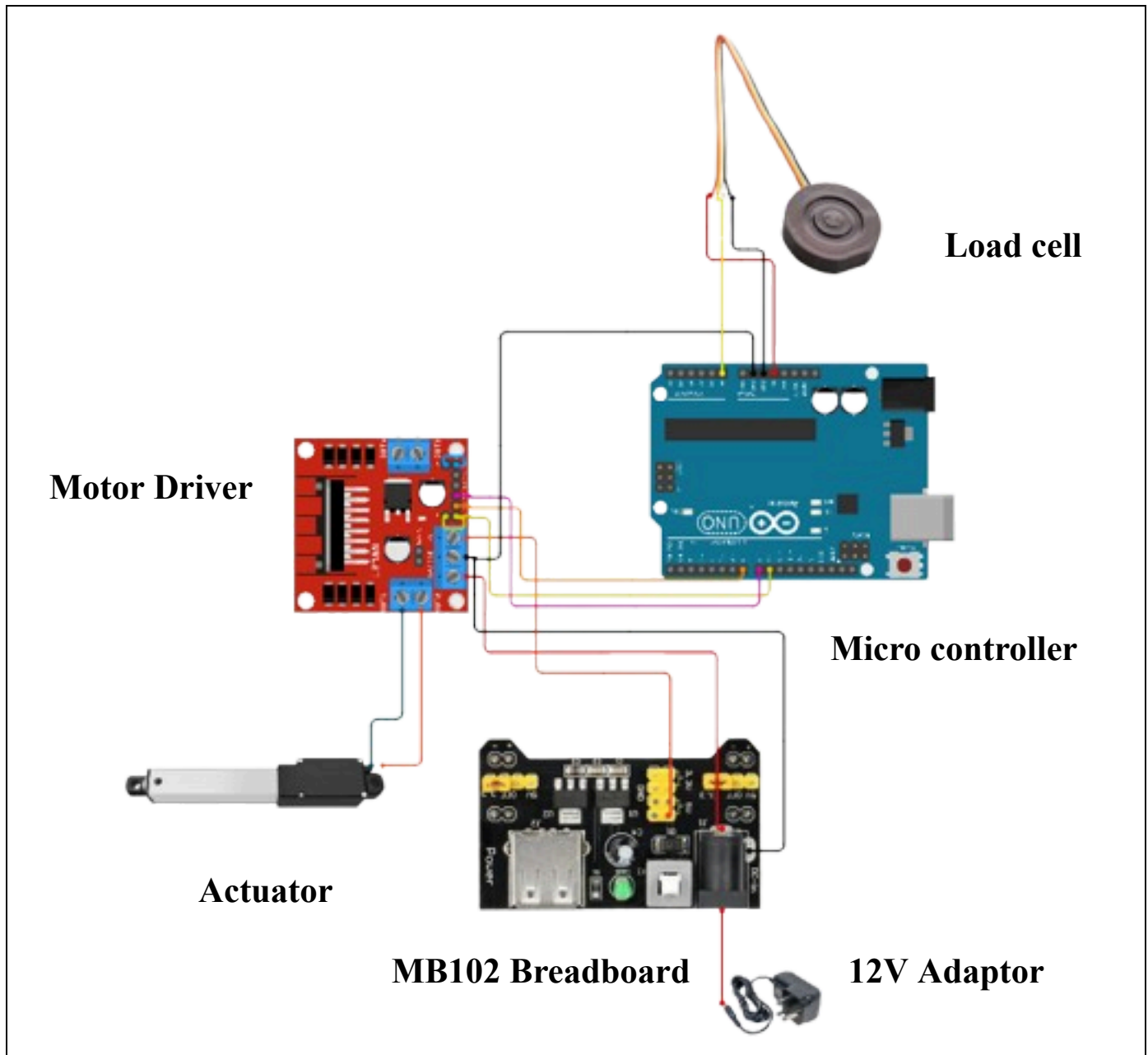
In our setup, we employ a feedback control system where the user specifies the desired value through an interface, and the actual output is measured by a load cell positioned beneath the actuator. Between the load cell and the actuator, we place the insole that needs to be calibrated. The actuator continues to extend until the load cell reading, which reflects the force applied to the insole, matches the user-defined setpoint. Achieving precise control of the motor, however, is challenging due to inherent system delays and errors. To address this, a PID (Proportional-Integral-Derivative) control system is implemented. The PID controller continuously adjusts the actuator's movement by minimizing the difference between the setpoint and the measured output, effectively reducing transient delays and

steady-state errors. This results in smoother, more accurate, and reliable system performance.



A PID (Proportional-Integral-Derivative) control system addresses transient delays and overshoots in non-ideal systems by minimizing errors between desired and actual outputs. The control law $u(t) = K_p \cdot e(t) + K_i \cdot \int e(t) dt + K_d \cdot de(t)/dt$ computes a corrective signal where $e(t)$ is the instantaneous error (setpoint minus measured output). The proportional term (K_p) adjusts responsiveness to current error, influencing transient behavior and steady-state error. The integral term (K_i) accumulates past errors to eliminate steady-state offsets, while the derivative term (K_d) predicts future error trends by differentiating consecutive errors, reducing oscillation magnitude and cycle duration. This computed $u(t)$ drives hardware (e.g., an Arduino outputting signals to an L298D motor driver), extending the shaft when $e(t) > 0$ (setpoint > output) and retracting it when $e(t) < 0$ (setpoint < output).

Electronic Circuit-



Components-

1. Arduino Uno
2. Actuonix L16 50mm Actuator
3. FX293X100A0050L Load cell
4. MB102 breadboard
5. L298D motor driver
6. 12V DC adaptor

Arduino Uno:

Arduino is a microcontroller commonly used to control a motor based on the output from a load cell, with the load cell's signal being fed back to the Arduino. In this setup, the load cell measures force or weight and sends its output to the Arduino, which processes the data and adjusts the motor's operation accordingly. This feedback loop allows for precise control, enabling the system to respond dynamically to changes in load, such as stopping the motor when a target weight is reached or adjusting speed for accuracy. This approach is widely used in automation, weighing, and material handling applications for its simplicity and effectiveness.

Actuator

An actuator is a crucial component that produces force or motion, typically by converting energy—such as electrical, hydraulic, or pneumatic—into mechanical movement.

Load cell:

The load cell serves as the primary element for achieving a feedback system in this setup. Once a set point is provided through the user interface, the output from the load cell is fed into the microcontroller, which continuously compares the measured value with the set point. Based on this error, the microcontroller drives the motor accordingly, adjusting parameters such as speed and direction. For instance, if the error is large, the motor operates at a higher speed to quickly reduce the difference. Additionally, if the error is

positive, the motor extends, and if the error is negative, the motor retracts. This closed-loop control ensures precise and dynamic adjustment, allowing the system to maintain the desired set point efficiently.

MB102 Breadboard:

The MB102 breadboard power supply module is a compact and versatile accessory designed to provide regulated power to standard breadboards used in prototyping and DIY electronics projects. It offers dual voltage outputs of 3.3V and 5V simultaneously, which can be selected via an onboard switch. The module can be powered through a micro USB port or a DC barrel jack with an input voltage range of 6.5V to 12V.

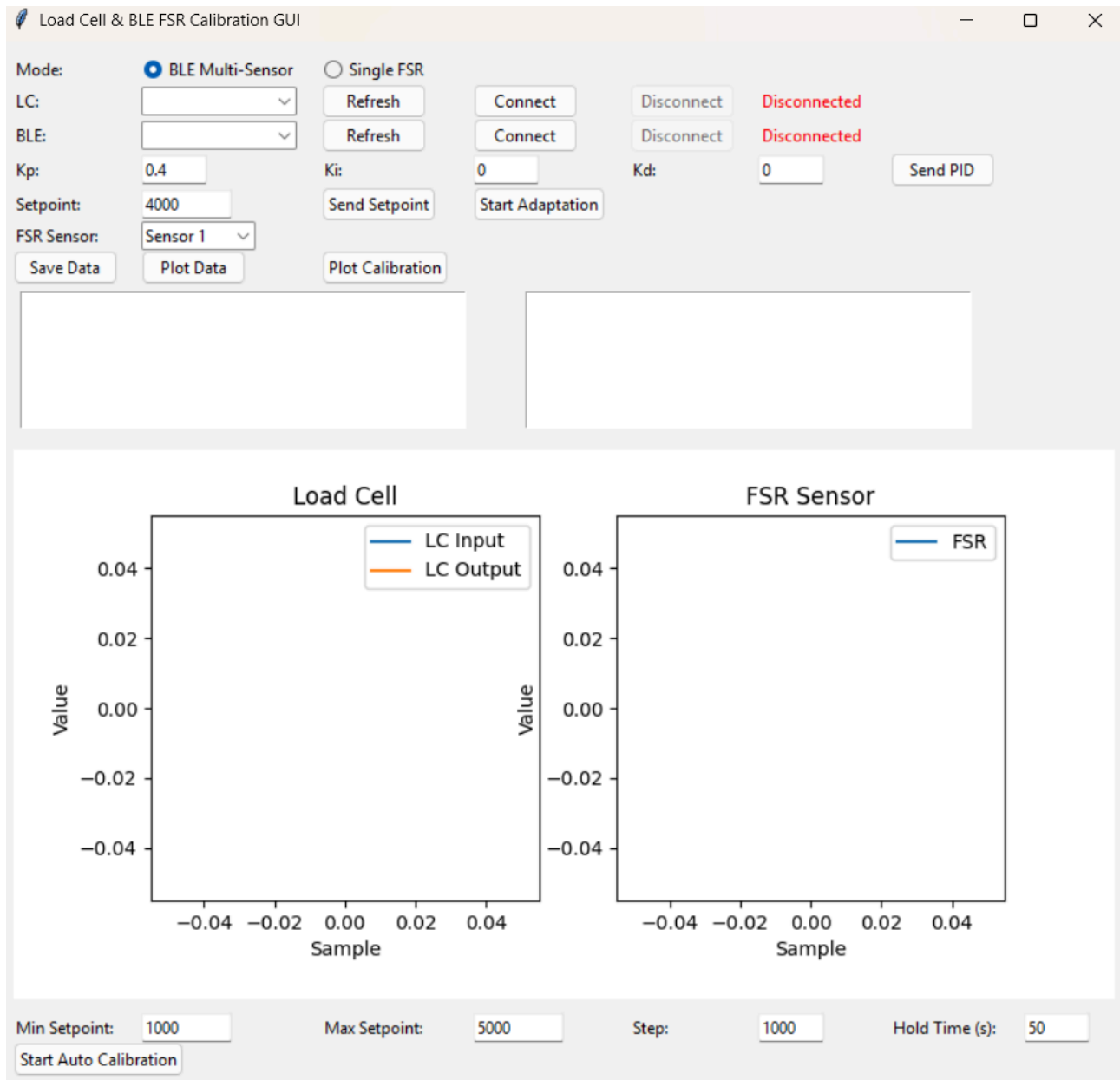
L298D Motor Driver:

The L298D motor driver is used to control both the speed and direction of a linear actuator. It is built around a dual H-Bridge driver chip, which enables independent control of two motors using standard logic signals from microcontrollers such as Arduino. In this application, the L298D module allows us to extend or retract the linear actuator by changing the direction of the motor, while also providing the ability to adjust the actuator's speed through pulse-width modulation (PWM). This makes the L298D an essential component for precise and flexible movement control in automation and robotics projects.

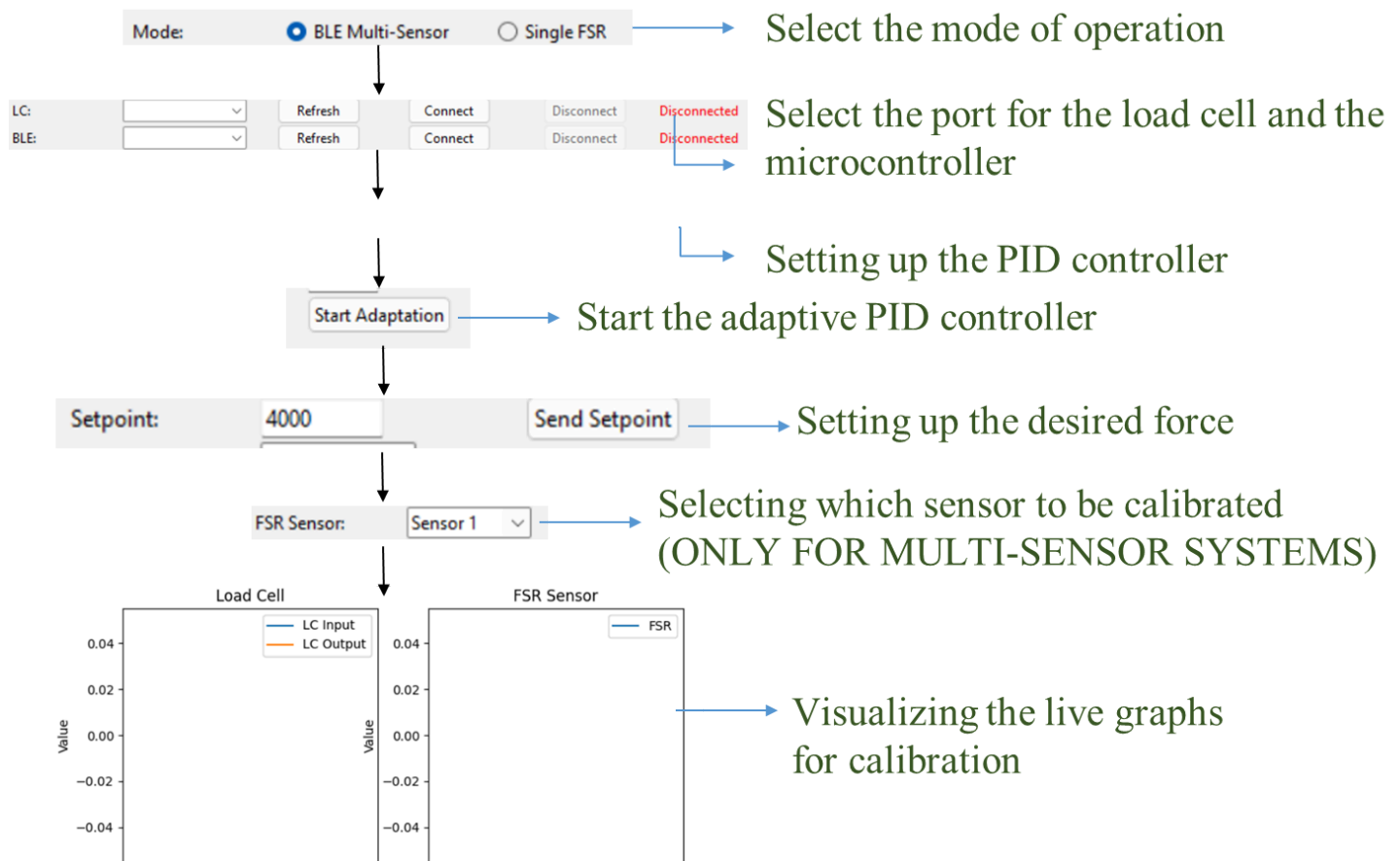
12V DC adaptor:

An adapter is used to power both the motor driver and the motor by providing the necessary voltage and current required for their operation.

User Interface-



Workflow of the UI-



Min Setpoint: 1000 Max Setpoint: 5000 Step: 1000 Hold Time (s): 50 → For auto calibration
Start Auto Calibration

Min Setpoint: 1000 Max Setpoint: 5000 Step: 1000 → Set the minimum setpoint, maximum setpoint, and resolution

Hold Time (s): 50 → Set the hold time for each step

Start Auto Calibration → Start auto calibration

Plot Calibration → For auto calibration, click this after end of operation

STEP BY STEP PROCEDURE OF OPERATION

1. Plug in the power supply to the SMPS (ensure power supply is OFF)
2. Connect the laptop with the UI, to the microcontroller
3. Open the UI and update all the settings as suggested in the WORKFLOW of the UI.
4. After setting up the UI switch on the Adapter, the motor should turn on applying the set point pressure
5. For every set point let the system work for 1 min after that save the data
6. Repeat the same procedure for the next set points
7. Save all the data's after every 1 min of operation for each setpoint
8. For auto calibration and operation of UTM steps are mentioned above in the workflow

Features-

Our Universal Testing Machine (UTM) is primarily designed to provide precise and controlled force for calibrating various force sensors, especially those used in healthcare applications. Unlike conventional UTMs, which are typically used to test the durability or mechanical properties of materials, our device is specialized for sensor calibration purposes. It enables the application of specific forces required to accurately calibrate sensors that operate based on force measurements. One key application in healthcare is the calibration of insole sensors, which are critical for measuring pressure and force distribution in medical diagnostics and rehabilitation. By focusing on controlled force application rather than material testing, our UTM offers a reliable and efficient solution tailored for sensor calibration needs in healthcare and other force-based sensing applications.

Conclusion-

In conclusion, the development of our compact Universal Testing Machine (UTM) marks a significant advancement in the field of affordable healthcare technology, particularly in the precise calibration of force sensors used in medical applications. Unlike traditional UTMs, which are primarily designed to test the mechanical strength and durability of materials, our system is purpose-built to deliver controlled, low-range forces tailored for sensor calibration. This is especially crucial in applications such as GAIT analysis, where accurate force measurement from insole sensors directly impacts patient diagnostics and rehabilitation outcomes. The robust mechanical design, featuring a high-stiffness acrylic structure, ensures minimal deformation and reliable force application up to 120N, while the integration of a feedback control system using a load cell and PID controller allows for real-time, precise adjustments to the actuator's movement and adaptability for various calibration scenarios. By focusing on sensor calibration rather than material testing, our UTM addresses a critical gap in healthcare technology, enabling the accurate and repeatable calibration of force-based sensors like those embedded in medical insoles. This not only improves the reliability of patient assessments but also supports broader efforts to make advanced medical tools more accessible and cost-effective. Ultimately, our innovation aligns with the mission of the Center of Excellence in Affordable Healthcare (CoE-AH) by bridging advanced technology and practical healthcare needs, fostering a future where high-quality medical diagnostics are within reach for all.